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EU AI Act

Spain

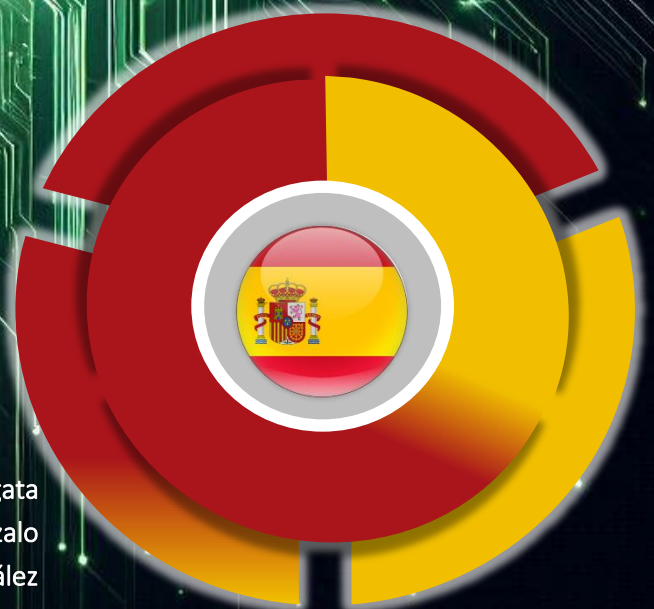
AI Ecosystem Brief

December 2025



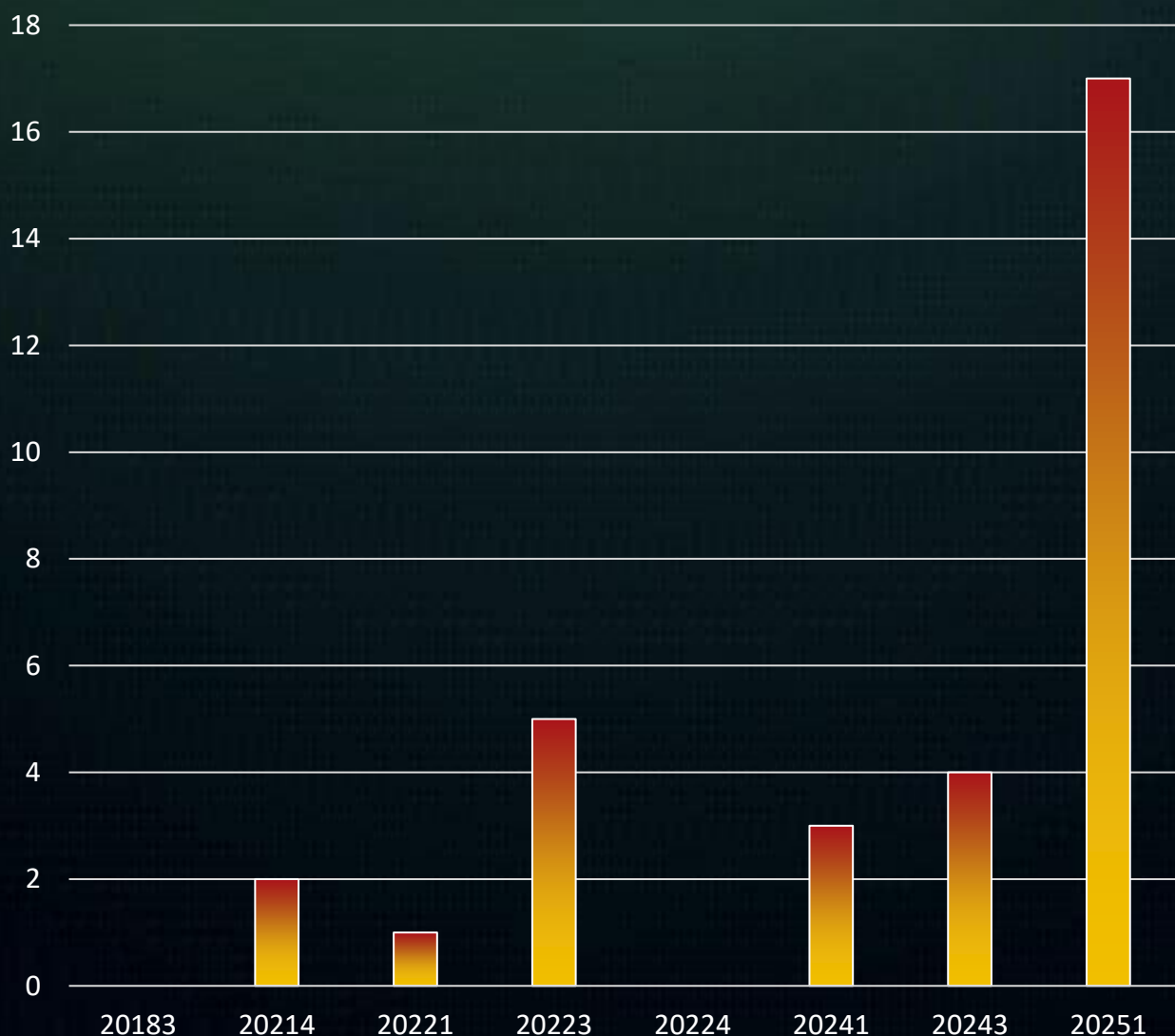
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Strengthening Spain's Generative AI Investment Trajectory: From Emerging Momentum to Scaled, Sustained Venture Capital Growth Through 2030

OECD.AI: Sum of VC investments (USD Millions) in generative AI in Spain (November 2025)



Explanation: This chart displays venture capital investments in generative AI in USD millions in Spain.

Spain's venture-capital investment activity in **generative AI** remains emergent, volatile, and highly sensitive to broader funding cycles. Based on available quarterly data (2018Q3–2025Q1), deal activity begins at **zero** in 2018 and remains minimal through early 2020, reflecting the nascent state of Spain's generative AI ecosystem and limited early-stage specialisation among domestic investors.

A modest uptick appears in **2021Q2 (1 deal)**, followed by a more visible rise in **2022**, peaking at **5 deals in 2022Q3** — a period coinciding with accelerating interest in foundational models globally, the first wave of generative AI commercialisation, and growing Spanish public-sector initiatives supporting data and AI experimentation. The temporary drop to **zero deals in 2022Q4** illustrates the sensitivity of Spain's market to global investment slowdowns and domestic constraints, including limited deep-tech financing capacity.

Investment re-accelerates in **2024**, with **3 deals in 2024Q1** and **4 in 2024Q3**, reflecting renewed investor appetite aligned with Spain's updated national AI strategy (ENIA 2024), growing HPC investment, and the emergence of new Spanish generative-AI startups targeting enterprise, language technologies, and sector-specific use cases.

A major spike occurs in **2025Q1**, with **17 VC deals**, representing by far the largest quarterly activity in the dataset. While this surge likely reflects a combination of maturing startups, improved availability of local and EU-level investment instruments, and spillover from global gen-AI financing, it also highlights the underlying **volatility** of Spain's ecosystem: rather than a stable upward curve, the trajectory is marked by sharp peaks and troughs depending on investor sentiment, pipeline maturity, and macroeconomic conditions.

This pattern points to a generative-AI investment landscape that is **accelerating but structurally fragile**. Spain shows the capacity for meaningful bursts of activity — driven by strong research assets, sectoral data spaces, digital-innovation programmes and regional initiatives (e.g., RETECH IA) — yet lacks the continuity associated with deeper, more specialised VC ecosystems. Constraints include: limited deep-tech funds with AI expertise, difficulty scaling seed-stage companies into growth rounds, and a still-emerging base of generative-AI startups with defensible IP and access to large training compute.

The data thus depict a system **in transition**: early signals of scaling and investor engagement exist, but the volatility indicates dependence on episodic factors rather than embedded structural capacity.

To stabilise and grow Spain's generative-AI investment environment, several strategic steps are advisable:

1. **Strengthen deep-tech and AI-specialist funding capacity:** Expand or co-fund dedicated generative-AI investment vehicles to provide continuity between seed and growth stages, reducing quarter-to-quarter fluctuations.
2. **Scale applied generative-AI commercialisation:** Link public AI infrastructure (HPC, data spaces, regulatory sandboxes) with investment pipelines so startups can more easily convert research into fundable, market-ready products.
3. **Broaden regional deal-flow development:** Use RETECH IA and regional technology centres to support generative-AI venture creation outside major hubs, increasing ecosystem resilience.
4. **Create investor-ready benchmarking and visibility tools:** Publish quarterly dashboards on generative-AI startups, models, datasets and applied pilots to reduce information asymmetry and attract international co-investment.
5. **Increase mid-stage capital availability:** Develop blended financing models with public guarantees to support Series A/B rounds, reducing dependence on global sentiment swings.

In summary, Spain's generative-AI VC activity shows clear upward momentum but pronounced instability, with sharp peaks (notably 2025Q1) punctuated by periods of very low activity.



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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

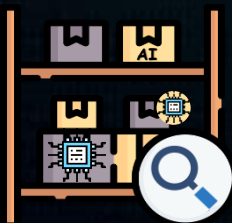
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contents

This section outlines the structure of the report, Spain's AI policy landscape, regulatory initiatives, governance mechanisms, findings, and recommendations. It serves as a navigation guide, enabling readers to quickly locate specific analyses, datasets, and policy discussions relevant to Spain's evolving artificial intelligence ecosystem.

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Overview

Artificial Intelligence (AI) is reshaping economies and governance worldwide, and Spain is establishing itself as a regional leader in ethical and human-centric AI development.

Anchored in the European Union's evolving regulatory landscape, Spain has adopted a supervisory-first approach that emphasizes institutional oversight, rights-based safeguards, and structured alignment with the EU AI Act. Rather than relying on sweeping national legislation, Spain is leveraging strategic frameworks, regulatory sandboxes, and international standards to ensure trustworthy and inclusive AI deployment.

At the heart of Spain's AI governance strategy is the Spanish Agency for the Supervision of Artificial Intelligence (AESIA)—Europe's first dedicated AI supervisory authority. AESIA will oversee enforcement, inspection, and awareness around AI systems in alignment with EU requirements. Complementing this are national instruments like the Royal Decree 817/2023, which introduced a regulatory sandbox for testing high-risk AI systems, and strategic resources including the National AI Strategy (2020–2025) and the Charter of Digital Rights, which articulate Spain's vision for safe, rights-respecting, and inclusive AI deployment.

Spain's AI governance model is supported by a robust institutional architecture. Beyond AESIA, entities like the Artificial Intelligence Advisory Council and the

Ministry of Digital Transformation contribute to shaping policy and long-term strategies. These bodies work across public, private, and civil society sectors to embed AI governance within national objectives such as digital transformation, social inclusion, and economic resilience.

Spain's approach also reflects strong commitment to international standards. The country is an active contributor to the OECD AI Principles, adheres to UNESCO's Recommendation on the Ethics of AI, and plays a central role in shaping EU-wide AI policy under the Digital Agenda for Europe and the Coordinated Plan on AI (2019–2027). Spain's participation in multilateral forums underscores its intent to harmonize domestic AI efforts with global norms and promote interoperable, human-centric governance.

As outlined in its National Artificial Intelligence Strategy, Spain aims to transform key sectors—such as healthcare, energy, tourism, education, and public services—through targeted AI deployment. Policy goals include fostering AI innovation through regional clusters and digital hubs, advancing data governance and compute infrastructure, and supporting the development of a future-ready workforce. Resources like the regulatory sandbox (RD 817/2023), strategic AI investment plans, and inclusive digital education programmes help operationalize Spain's governance commitments.

Through a combination of EU-aligned oversight, structured policy frameworks, and international engagement, Spain is building a comprehensive and adaptive model of AI governance. By prioritizing ethics, transparency, and societal trust, Spain is charting a course toward AI that is both innovative and democratically accountable.



Specific AI

Governance or Law

Rather than enacting separate AI laws, Spain aligns with the EU AI Act, establishing AESIA, Europe's first dedicated AI supervisory agency. Spain supports responsible AI via the RD Sandbox for testing high-risk AI, alongside non-binding frameworks like the National AI Strategy and Digital Rights Charter promoting ethical, rights-respecting AI development.

Specific AI Governance in Spain

Rather than enacting its own sweeping AI legislation, Spain has opted to align closely with the European Union's regulatory framework—most notably the EU AI Act—while laying national groundwork to implement and supervise its provisions effectively. Central to this strategy is the establishment of the Spanish Agency for the Supervision of Artificial Intelligence (AESIA), Europe's first dedicated AI supervisory body. AESIA is tasked with a broad remit including oversight, inspection, and enforcement related to AI systems, in line with EU-level requirements. Spain's commitment to responsible AI is operationalized through instruments such as the Royal Decree 817/2023 ("RD Sandbox"), which sets up a controlled testing environment to support organizations in trialing high-risk AI systems under real-world conditions. The aim is to produce guidance and best practices to support future compliance with the EU AI Act. Complementing this are non-binding frameworks like the National AI Strategy and the Charter of Digital Rights, which articulate Spain's vision for ethical, rights-respecting, and inclusive AI development.

AI Ethics Guidelines and Sectoral Governance Tools

In the absence of AI-specific national legislation, Spain relies on a layered governance model grounded in the EU AI Act and supplemented by existing domestic laws and policy instruments. While not directly regulating AI, a range of Spanish statutes—including those governing data protection, competition, labor rights, audiovisual communication, and consumer protection—have implications for AI system deployment. Guidance frameworks such as the National AI Strategy (2020–2025) provide policy direction across sectors, with emphasis on transparency, accountability, and fundamental rights. Meanwhile, advisory bodies like the Artificial Intelligence Advisory Council offer expert input to the Ministry of Digital Transformation, promoting debate and insight on evolving AI issues. Spain also adheres to international principles for AI governance through its participation in organizations such as the OECD and the Council of Europe, reinforcing its commitment to global AI standards.

Proposed AI Regulatory Authority and Future Outlook

Spain's AI oversight architecture is anchored by AESIA, which will play a central role in supervising AI implementation once the EU AI Act comes into force. AESIA's mandate includes not only enforcement and sanctioning but also promoting awareness, training, and safe AI adoption across the public and private sectors. Looking ahead, the results of the RD Sandbox initiative are expected to shape future national regulatory instruments, ensuring Spain's legal framework remains aligned with both domestic needs and European standards. The country's structured, EU-coordinated approach aims to maximize AI's economic and social benefits while ensuring it remains trustworthy, inclusive, and compliant with core democratic values.



Spanish National AI Strategy

Spain's AI governance approach is grounded in a value-driven, EU-aligned framework that combines ethical safeguards, public sector leadership, and technological innovation to promote responsible AI deployment. Through instruments like the National AI Strategy, the Digital Rights Charter, and emerging regulatory mechanisms such as AESIA and the AI Sandbox, Spain fosters context-sensitive, inclusive, and trustworthy AI adoption.

Spain's AI governance framework is anchored in values of responsibility, human-centricity, transparency, and strategic sovereignty. These guiding principles are translated into action through its Estrategia Nacional de Inteligencia Artificial (ENIA), most recently updated in May 2024. The revision strengthens Spain's high-performance computing (HPC) capabilities and explicitly addresses the rapid rise of generative AI across sectors.

Unlike purely reactive compliance frameworks, Spain's strategy is both visionary and institutionally robust. It embeds monitoring, reporting, and evaluation mechanisms: an independent Advisory Council on Artificial Intelligence supports policymakers in defining initiatives, while explicit key performance indicators (KPIs) are assigned to each programme.

Most initiatives are tracked via internal dashboards, and annual progress reports are submitted to the Council of Ministers.

A standout feature of Spain's approach is its regulatory infrastructure. Spain was the first EU country to establish a dedicated AI supervision agency, the Agencia Española de Supervisión de la Inteligencia Artificial (AESIA), tasked with ensuring compliance with EU law and increasing public understanding of AI's societal impact. To foster innovation while maintaining oversight, Spain also pioneered AI regulatory sandboxes — a pilot programme launched in December 2024 to test up to 12 high-risk AI systems, especially from SMEs and start-ups, and co-develop technical guidelines ahead of broader regulation.

Spain strongly emphasizes building its AI innovation ecosystem. Through the Territorial Networks of Technological Specialization (RETECH IA), the country coordinates across all its autonomous regions. Thirteen flagship projects have been launched, covering areas such as tourism (Spain Living Lab), industry value chains, smart rural infrastructure, digital health, and even cultural heritage. These regional networks foster cross-sector public-private partnerships and drive local AI capacity building.

Investment in research and talent is central to Spain's strategy. Among its initiatives are the ENIA Chairs (university-business chairs in ethics, AI and health, green algorithms, etc.), and fifteen interdisciplinary research networks focusing on areas such as cognition, algorithms, social challenges, and quantum.

Spain also supports talent development through scholarships and grants: for example, a programme offering 374 four-year scholarships in AI and related fields was launched in 2024. There is also Forma Inteligencia Artificial, which subsidises up to 80% of course fees for nearly 8,000 students in AI-related university programs.

Spain's strategy adopts a risk-sensitive, multi-domain approach, targeting key sectors where AI can have social and economic impact. In healthcare, Spain is building a national health data space to support AI-driven diagnostics and secondary use of health data, while working to bridge regional fragmentation in data governance.

For the environment, Spain has launched the National Green Algorithms Program (PNAV), promoting "green-by-design" AI and energy-efficient infrastructure, including certifications for AI energy consumption.

It has also initiated a large consortium, IA4TES, to apply AI to the sustainable energy transition. Partnerships and international collaboration are also

foundational. Spain has signed a Memorandum of Understanding with IBM to develop Spanish-language foundational AI models (both large and small), reinforcing linguistic sovereignty and ensuring generative AI models reflect Spanish (and co-official) languages.

Digital Government

From a societal inclusion perspective, Spain places strong emphasis on AI literacy, upskilling, and equitable access. The newly created AESIA is running a national AI literacy plan, focused on citizens, businesses (especially SMEs and startups), public administrations, and education systems.

Digital Strategy

Meanwhile, the national skills strategy includes workforce transition programmes, technical university degrees, and collaboration with private actors to ensure broad participation in AI capacity building.

Finally, Spain's AI strategy is designed to be adaptive and future-oriented. The 2024 update reflects a recognition of accelerating technological change—especially in generative AI—and signals a willingness to revise priorities as needed. The alignment with the EU AI Act, via AESIA and the sandbox, also indicates Spain is building a regulatory model that is simultaneously compliant, innovative, and participatory.

Promote scientific research, technological development and innovation in AI.

Promote digital capabilities, empower national talent and attract global skills in the field of AI.

Develop data platforms and technological infrastructures in support of AI.

Incorporate ai into value chains to transform the economic fabric.

Enhance the use of ai in government administration and in national strategic missions.

Establish an ethical and regulatory framework that reinforces the protection of individual and collective rights, in order to guarantee inclusion and social welfare.

Approach to AI Regulation

Spain has positioned itself as a proactive participant in shaping AI governance within the European Union, establishing dedicated institutions and test environments while aligning closely with EU legislation. While not enacting standalone AI laws, Spain fosters responsible AI development through strategic initiatives, ethical principles, and regulatory innovation aimed at building a robust, human-centric AI ecosystem.

1. Regulatory and Strategic Frameworks

Spain's approach to AI governance is rooted in alignment with EU initiatives, particularly the forthcoming EU AI Act, rather than standalone national legislation. Although Spain has not introduced binding national AI regulations, it has laid down a strategic foundation through non-binding instruments such as the Charter of Digital Rights and the National Artificial Intelligence Strategy (2020–2025). These frameworks embed ethical and social values into AI development, emphasizing human agency, transparency, and accountability. Additionally, Spain's Royal Decree 817/2023 created a regulatory sandbox aligned with the EU AI Act, offering a controlled environment for piloting high-risk AI systems. This regulatory innovation supports compliance testing and iterative learning, enabling organizations to adapt to EU requirements efficiently.

2. Institutional Support and Strategic Missions

Institutional leadership is spearheaded by the newly established Spanish Agency for the Supervision of Artificial Intelligence (AESIA), the first dedicated AI regulatory body in Europe. AESIA is tasked with monitoring, inspection, and sanctioning powers under the EU AI Act, as well as promoting awareness, training, and collaboration around AI governance. Complementing AESIA is the Artificial Intelligence Advisory Council, which advises the Ministry of Digital Transformation on public policy and long-term AI strategies. Together, these institutions represent a coordinated effort to operationalize Spain's AI ambitions across government, industry, and civil society.

3. Standards Development and Global Engagement

Spain is actively engaged in international AI standard-setting and ethics debates, including its participation in OECD AI principles and European Commission-led initiatives. The National AI Strategy underscores Spain's intent to harmonize domestic efforts with international norms through forums such as the Council of Europe and OECD, as well as through adherence to EU-wide digital strategies like the Digital Agenda for Europe and the Coordinated Plan on AI (2019–2027). By contributing to EU policy formulation and regulatory experimentation, Spain positions itself as a bridge between domestic innovation and supranational coherence in AI governance.

4. Sectoral and Extended Regulatory Approaches

Spain does not mandate sector-specific AI laws but anticipates sectoral implementation to evolve under the umbrella of the EU AI Act. The RD Sandbox allows high-risk AI deployments in real-world settings, promoting sector-specific insights and risk-mitigation practices. This framework facilitates AI experimentation in areas such as healthcare, finance, and labor, with potential future applications extending across education, public administration, and media. Spain's broader legal landscape—including data protection, labor law, competition law, and digital commerce—is already impacting AI indirectly, setting legal guardrails around AI applications.

5. Fostering Innovation and Indigenous Development

Innovation is central to Spain's AI agenda, driven by policies that aim to nurture domestic AI capabilities and ensure economic competitiveness. The regulatory sandbox provides a unique environment for Spanish enterprises to develop, test,

and iterate AI solutions in compliance with EU norms. Public-private partnerships and the promotion of AI ethics in development signal Spain's commitment to fostering a responsible and competitive AI ecosystem. Government support for open experimentation and local R&D reflects a long-term strategy to build sovereign AI capacity.

6. AI in Public Sector Transformation

While Spain's AI governance is still evolving, the state has begun integrating AI into digital public services and public sector innovation. AESIA is expected to play a central role in guiding AI integration across government systems, ensuring legal compliance and ethical deployment. The Charter of Digital Rights provides a normative foundation for public AI adoption, advocating for citizen-centricity, accountability, and equitable digital transformation.

7. AI Skills, Education and Inclusion

Spain's National AI Strategy includes provisions for developing AI-related skills and enhancing societal readiness for technological change. It places strong emphasis on inclusive education, research support, and talent development as pillars of responsible AI adoption. While no single national programme mirrors Singapore's SkillsFuture, Spain integrates AI education into broader digital literacy efforts, and emphasizes social cohesion and fundamental rights protection in the digital transition.

8. Infrastructure, Compute and Data Foundations

Though still emerging, Spain is gradually building the data and infrastructure foundations necessary for scalable AI deployment. The RD Sandbox enables secure data testing in real-world conditions, while public policies are beginning to address compute access and secure data sharing frameworks.



Approach to Foster Innovation

Spain's comprehensive approach to AI governance, deployment, and innovation is rooted in EU alignment, ethical safeguards, and institutional coordination. The National AI Strategy (ENIA) leverages AI as a catalyst for social cohesion, economic competitiveness, and public sector transformation, while embedding trust and inclusivity at the core of technological progress.

1. Unlocking Economic Value through Targeted AI Deployment

Spain's National AI Strategy (ENIA) aims to transform the productive fabric of the economy by integrating AI across critical sectors such as healthcare, tourism, agriculture, education, energy, and mobility. AI is viewed as a lever to boost competitiveness, productivity, and sustainability. Early deployment is promoted in areas where Spain can generate significant societal value, particularly in strategic missions with AI enhancing service delivery.

2. Catalysing Innovation through Strategic Clustering and Ecosystems

Spain supports AI innovation through regional hubs and public-private collaboration. The Strategy envisions a national ecosystem anchored by research centres, digital innovation hubs (DIHs), and universities. A forthcoming Network of Excellence in AI will connect interdisciplinary research entities with government and industry. This mirrors Singapore's clustering efforts and helps embed AI in regional economies, from agriculture in rural areas to biotech in urban clusters

.3. Enabling Trusted and Responsible Data Use

Spain prioritizes data governance and interoperability as foundational to AI development. The Strategy promotes secure data sharing, development of open-source platforms, and compliance with EU data protection laws. AI-related data infrastructure, including high-performance computing and general-purpose AI platforms, are being expanded to support innovation while upholding privacy, integrity, and legality

.4. Scaling AI in the Public Sector

Public administration is both a user and enabler of AI under ENIA. Applications span sectors such as justice, environment, health, education, and transportation. The Strategy advocates AI adoption in service design, public planning, and citizen interaction to boost transparency and responsiveness. Government-led deployment demonstrates use-case viability and drives citizen trust in AI systems

.5. Advancing Compute and Research Infrastructure

Spain's investment in high-performance computing, AI data platforms, and open digital infrastructures underpins its AI ambitions. The country's strong university system and public R&D investment support both foundational and applied AI research. Spain's scientific contributions in areas like natural language processing, robotics, and energy optimization are recognized internationally

.6. Empowering SMEs and Startups

ENIA emphasizes the digital transformation of SMEs as a national priority. Public innovation policies—including funding for experimentation and access to DIHs—are tailored to help smaller firms adopt AI solutions. Public-private consortia, startup participation in AI research projects, and support for spin-offs strengthen the competitiveness of Spain's SME sector

.7. Developing a Future-Ready Workforce

The Strategy focuses on enhancing digital and AI-related capabilities across the population. Interdisciplinary training programs promote skills in both STEM and social sciences. Gender equality, digital literacy, and inclusion of underrepresented groups are central to capacity-building initiatives. AI education is integrated into lifelong learning and university systems to ensure a robust talent pipeline

.8. Aligning AI Innovation with Sustainability Goals

Spain leverages AI to accelerate ecological transition and support the European Green Deal. AI applications in climate forecasting, energy efficiency, and circular economy are promoted. The Strategy emphasizes the development of energy-efficient AI systems and encourages the use of AI to monitor and reduce environmental impact, reflecting Spain's dual commitment to innovation and sustainability

.9. Building Public Trust and Ethical AI

Trust is a cornerstone of Spain's AI framework. The Digital Rights Charter, ethical oversight bodies, and transparency mechanisms ensure that AI aligns with democratic values and citizen rights. Public participation in policy design and ethical deliberation ensures broad societal input. Spain's approach supports AI that is explainable, accountable, and aligned with social welfare.

10. Shaping Global AI Governance and Standards

Spain is an active contributor to EU-wide AI regulation and global frameworks including those from the OECD and UNESCO. The Strategy reinforces Spain's commitment to international collaboration and the development of interoperable, human-centric AI standards. Through EU initiatives and multilateral forums, Spain seeks to shape a global AI order rooted in inclusion, security, and ethical integrity



Digital Decade

2025: Report

As of 2024, approximately 11.31% of enterprises in Spain were using artificial intelligence (AI) technologies. This figure is slightly below the EU average of 13.48%, but it aligns with Spain's national trajectory. From 2023 (9.18%) to 2024, Spain saw a 23.2% increase in AI adoption, which remains well below the EU's average growth rate of 67.2%. A closer look shows disparities in adoption: 10.3% of SMEs use AI, compared to 43.96% of large enterprises—a gap of 33.66 percentage points, larger than the EU average of 28.53 points.

To boost this trajectory, Spain launched HispanIA 2040 in January 2025—its long-term strategic vision for how AI can shape the country's future. At the same event, Spain introduced ALIA, the country's first foundational AI model trained on extensive data in Spanish and co-official languages (Catalan, Galician, and Basque). ALIA is designed as an open-source, publicly managed infrastructure to reinforce technological sovereignty and offer a transparent alternative to dominant English-language models. Spain is also building an open-source community around ALIA, encouraging collaboration, reuse, and widespread adoption across both public and private sectors. In a major regulatory development, Spain launched a regulatory sandbox in April 2025 for high-risk AI systems. This sandbox began by testing 12 systems for compliance with the European AI Regulation, including aspects such as human oversight.

It aims to help startups and SMEs adapt to the upcoming EU AI Act, set to take effect in August 2026.

These strategic actions are designed to reduce adoption barriers, foster innovation, and ensure broad access to the benefits of AI throughout the economy. Spain was advised to continue supporting the digitalisation of businesses, with a special focus on fostering the uptake of advanced technologies.

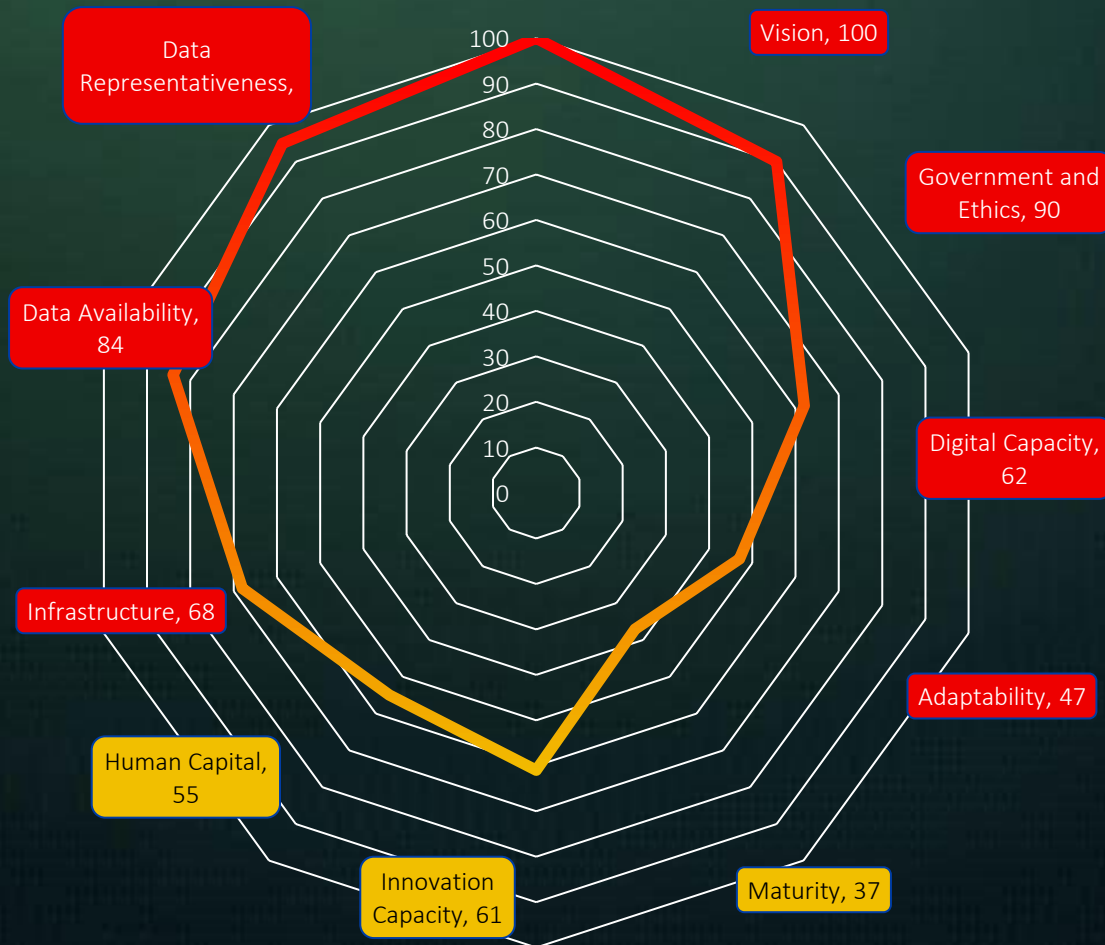
In response, Spain implemented policy actions in 2024, achieving a 4.8% increase in the basic digital intensity of enterprises over two years. Adoption of advanced technologies grew by 11.61% compared to 2023, reflecting the impact of various national programmes. Nevertheless, challenges remain, particularly with AI adoption still below the EU average (11.3% vs 13.5%) and the large adoption gap between SMEs and large enterprises.



Government AI Readiness

Western Europe ranks as the highest-performing region in this year's Government AI Readiness Index, with five countries in the global top 10. France leads the regional ranking with a score of 79.36, narrowly ahead of the United Kingdom (78.88), followed closely by the Netherlands (77.23), Germany (76.90), and Finland (76.48). With an average regional score of 69.56, Western Europe outperforms the global average (47.59) across all three pillars of Government, Technology, and Data & Infrastructure. The region's standout strength is the Data & Infrastructure pillar, where it averages 81.91, more than 21 points above the global benchmark (59.62), reflecting its advanced digital infrastructure and high-quality data environment. Although a few microstates such as San Marino, Liechtenstein, and Andorra fall slightly below the global average in the Government pillar, Western Europe's dominance at the top of the index underscores its continued leadership in AI readiness.

2024 Scores by Dimension



Government

Spain scores 74.58 in the Government pillar, reflecting its proactive institutional efforts to prepare for the EU AI Act and position itself as a leader in AI regulation within Europe. The country's approval of the Royal Decree 729/2023, establishing the Spanish Agency for the Supervision of Artificial Intelligence (AESIA), marks a pioneering step in AI oversight. Spain also introduced the Royal Decree 817/2023, creating a regulatory sandbox aligned with EU guidelines for high-risk AI systems. These initiatives, along with Spain's National AI Strategy (2020–2025) and adherence to the Charter of Digital Rights, demonstrate a comprehensive governance framework focused on ethical AI, public trust, and coordinated digital transformation. The government's multilevel coordination—spanning national, regional, and EU dimensions—enhances legal harmonization and forward-looking policy design in AI governance.

Technology Sector

Spain's Technology pillar score of 50.75 indicates a developing but uneven AI innovation landscape. While the country hosts several research institutions and participates actively in EU-wide AI initiatives, its domestic AI ecosystem still faces challenges related to commercialization, private-sector investment, and the scalability of start-ups. Notable programs such as the Artificial Intelligence Advisory Council and public-private collaborations provide strategic direction, but deeper integration with industrial R\&D and entrepreneurial acceleration is needed. Spain's emerging AI clusters and regional technology hubs offer promise, particularly in sectors like language technologies, health, and energy, yet further investment in talent retention, research infrastructure, and high-growth tech ventures is essential to close the innovation gap with higher-scoring European peers..

Data & infrastructure Sector

Scoring 82.43 in the Data & Infrastructure pillar, Spain demonstrates significant strength in digital readiness and data ecosystem development. Its robust telecommunications framework, extensive broadband coverage, and growing adoption of cloud services support scalable AI experimentation. Government initiatives around secure data sharing, digital rights, and cross-border interoperability signal a commitment to data governance aligned with EU norms. Spain's active participation in initiatives such as the European Data Strategy and GAIA-X reinforces its dedication to fostering a trusted and interconnected data environment. Combined with national efforts to promote open data and digital literacy, these elements create a strong foundation for AI deployment and data-driven innovation across sectors.

With a total score of 69.25, Spain performs close to the Western European regional average (69.56) in the Government AI Readiness Index. Its clear regulatory ambition, progressive infrastructure base, and alignment with EU AI governance frameworks position the country as an emerging leader in AI oversight and ethical implementation. While Spain's technology sector lags behind regional frontrunners, strategic investments, institutional coordination, and sandbox experimentation suggest promising momentum toward a more mature and resilient AI ecosystem.

Conclusion

Spain's artificial intelligence ecosystem is navigating a transformative phase—anchored in ethical governance and EU alignment, with growing momentum toward innovation at scale. With strong institutional commitments, including the establishment of AESIA (the first AI supervisory body in Europe) and strategic tools like the National AI Strategy (ENIA) and the Royal Decree 817/2023 sandbox, Spain has laid a structured foundation for responsible AI deployment that integrates social values with economic opportunity.

Spain scores highly on data availability and digital infrastructure, thanks to national investments and EU-backed initiatives like GAIA-X and the European Data Strategy. This positions the country to scale AI deployment across sectors such as healthcare, public administration, and education, where early use cases already show promise. Spain's governance model—deeply rooted in rights-based, inclusive innovation—demonstrates a commitment to building AI systems that are explainable, accountable, and aligned with democratic norms.

Yet key challenges must be addressed. Spain's relatively modest technology sector score underscores the need to bridge gaps in AI commercialisation, research translation, and industrial integration. Strong academic foundations and public-private initiatives have yet to yield broad-scale entrepreneurial acceleration. Strengthening SME participation, deepening public R&D-private

sector linkages, and expanding high-performance compute infrastructure will be critical to realising Spain's full AI potential.

To sustain its strategic trajectory, Spain must intensify support for indigenous AI capabilities, particularly in regions outside major urban centres. A sharper focus on high-impact talent development, responsible data use, and regional ecosystem coordination will reinforce national resilience. Internationally, Spain's active role in the EU, OECD, and Council of Europe gives it a platform to shape global AI norms—especially in aligning AI with human dignity.

If met with continued institutional coordination, targeted investment, and civic trust, Spain can emerge not only as a regulatory frontrunner in Europe—but also as a model for dynamic, ethical, and socially cohesive AI leadership. The vision is clear—the opportunity to act decisively is now

Appendix A: OECD AI Progress Report

This note summarises the OECD country note on **Spain's** progress implementing its national AI strategy (“Estrategia Nacional de Inteligencia Artificial, ENIA” — updated in May 2024), and how those commitments align with broader national efforts around data, research, governance, and digital transformation.

Spain's AI strategy is embedded in a cross-sectoral vision, underpinned by its 2024 strategy update, which emphasises strengthening high-performance computing (HPC), embracing generative AI, and promoting human-centred, responsible AI. Institutional structures have matured: a dedicated supervisory agency (AESIA) has been established to enforce AI regulation, and an Advisory Council on Artificial Intelligence helps steer and monitor policy.

On the infrastructure and data side, Spain is investing heavily: its national AI plan channels significant funding to upgrade HPC (e.g., supercomputing capacity) to support large language model (LLM) training, with a new GPU-based supercomputer under development. It is also developing a “Plan for the Promotion of Sectorial Data Spaces” to promote secure data sharing and reuse.

In research and innovation, Spain actively supports public-private partnerships, interdisciplinary consortia, and SMEs. An example is the AI4ES network, which links leading technology centres to promote data- and AI-based innovation. Another key instrument is the use of **AI regulatory sandboxes**, launched to test

high-risk systems in collaboration with competent authorities and companies, guiding future compliance with the EU AI Act. Territorial cohesion is central to Spain's deployment strategy: through the RETECH IA initiative (Territorial Networks of Technological Specialization), 13 regional projects across all autonomous communities are underway — ranging from smart tourism, AI in industrial value chains, to rural data spaces.

On skills and talent, Spain's strategy is complemented by its National Plan for Digital Skills and other programmes (e.g., the “D Generation Pact”), aiming to train, attract, and retain AI talent. Meanwhile, in the public sector, there is a clear push to promote AI adoption — especially in administration, as well as in key sectors such as tourism, environment, and healthcare. In healthcare, Spain is building a national health data space (via its Digital Health Strategy), backed by funding, to enable interoperability, AI-driven diagnostics, and secondary use of health data.

Governance and accountability are also strengthening: Spain now regularly defines KPIs for each AI initiative, monitors progress via internal dashboards, and reports to the Council of Ministers. The supervisory agency AESIA is tasked with enforcing compliance and raising public awareness on AI's societal impacts.

Strengths

- **Updated strategic vision:** The May 2024 revision of ENIA aligns Spain's AI policy with generative AI trends, stronger HPC needs, and a focus on societal and sectoral impacts.
- **Dedicated regulation and oversight:** The creation of AESIA gives Spain a clear institutional mechanism for AI supervision and compliance.
- **Robust infrastructure investment:** Significant public funds are being channelled into HPC and supercomputing, including a next-generation GPU supercomputer and upgraded CPU-based systems.
- **Data ecosystem development:** Spain's plan to build sectoral data spaces reflects ambition to unlock data reuse across domains.
- **Innovation, SMEs & territorial cohesion:** Through public-private networks (e.g., AI4ES) and regional specialization (RETECH IA), Spain is leveraging distributed innovation capacity.
- **Governance and accountability:** The use of KPIs, reporting mechanisms, and advisory bodies strengthens oversight and alignment.

- **People-centred AI and skills:** By embedding AI in national skills plans and digital transformation policies, Spain is investing in its human capital.

Constraints and Risks

- **Long-term sustainability:** While the 2024 strategy has strong funding, the report suggests that investment plans (e.g., for HPC) must be updated beyond the immediate recovery era; the existing HPC investment plan expired in 2024.
- **Data governance complexity:** Establishing sectoral data spaces is ambitious, but implementing them securely and equitably across regions and sectors may be challenging, especially given Spain's decentralized structure.
- **Regulatory sandbox scaling:** The pilot regulatory sandbox is a valuable tool, but its initial scale (12 systems) may limit learning, and more capacity may be needed to absorb a broad variety of use-cases.
- **Talent and skills gap:** Even with national-level talent initiatives, the fast pace of AI development (especially generative AI) may outstrip the pipeline of skilled professionals.
- **Regional inequality:** Despite RETECH, regional disparities in digital infrastructure, AI capacity, and institutional maturity may slow coherent national adoption.
- **Transparency & public trust:** As AI systems are more deeply embedded (in public administration, health, tourism, etc.), ensuring transparency and meaningful public engagement will be critical.

Practical Recommendations

1. **Embed medium-term funding for HPC:** Develop a new multi-year investment plan (2025–2030) for HPC, aligning with generative AI needs, and secure budget continuity beyond RRF-era allocations.
2. **Accelerate operationalising data spaces:** Turn the Plan for Sectorial Data Spaces into concrete pilots with standardized legal frameworks, APIs, and regional data governance bodies; use RETECH networks to coordinate cross-community implementation.
3. **Scale regulatory sandbox efforts:** Expand the pilot to include more high-risk AI systems; create structured feedback channels to feed into AESIA's regulatory policies; share learnings publicly to build trust.

4. **Strengthen talent pipelines:** Grow initiatives under the Digital Skills Plan and D Generation Pact; establish fellowships or AI research chairs in universities; use public procurement to absorb trained talent into government AI projects.
5. **Enhance transparency and engagement:** Publish regular (annual) public reports on AI strategy KPIs, sandbox outcomes, and enforcement actions; conduct citizen dialogues on AI ethics, especially around public-sector deployment (e.g., in healthcare, administration).

Overall Assessment: The OECD country note portrays **Spain** as a country that is rapidly maturing its AI ecosystem: it has updated its national AI strategy to reflect generative AI, backed by strong institutional reforms (such as AESIA), significant infrastructure investment (HPC), and forward-looking innovation models (regulatory sandboxes, regional networks). However, to convert these foundations into sustained impact, Spain will need to secure long-term funding, operationalise its data-sharing ambitions, broaden regulatory experimentation, nurture talent at scale, and maintain transparency to build public trust.



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This section compiles all primary and secondary sources cited—legal texts, policy papers, academic studies, and official reports. It provides a transparent basis for the report’s analyses and recommendations, enabling verification, further research, and continued dialogue on AI policy and governance in Spain.

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